

Lab 3 - Routing Protocol

NAME: Evelyn Goh Yuan Qi

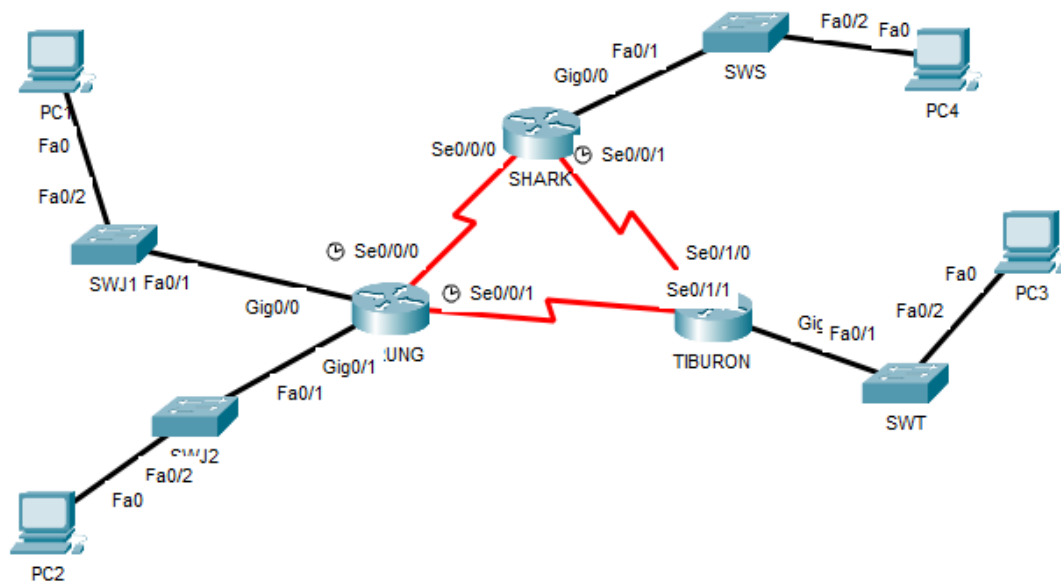
MATRIC NUM.: A23CS0222

SECTION: 12

INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

TOPOLOGY



LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

Table 1

Device	Subnetwork	Usable Hosts
JERUNG	LAN1	20
	LAN2	30
SHARK	LAN	60
TIBURON	LAN	50
Connections	JERUNG-SHARK	2
	JERUNG-TIBURON	2
	SHARK-TIBURON	2

Table 2

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	-
2		Se0/0/1	172.18.110.197	255.255.255.252	-
3		G0/0	172.18.110.1	255.255.255.224	-
4		G0/1	172.18.110.33	255.255.255.224	-
5	TIBURON	Se0/1/1	172.18.110.198	255.255.255.252	-
6		Se0/1/0	172.18.110.202	255.255.255.252	-
7		G0/0	172.18.110.129	255.255.255.192	-
8	SHARK	Se0/0/0	172.18.110.194	255.255.255.252	-
9		Se0/0/1	172.18.110.201	255.255.255.252	-
10		G0/0	172.18.110.65	255.255.255.192	-
11	PC1	-	172.18.110.30	255.255.255.224	172.18.110.1
12	PC2	-	172.18.110.62	255.255.255.224	172.18.110.33
13	PC3	-	172.18.110.190	255.255.255.192	172.18.110.129
14	PC4	-	172.18.110.126	255.255.255.192	172.18.110.65

LAB TASKS

Task 1 – IP Addressing

1. Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show your workings here and complete Table 2 with the correct information. PCs will be given the last usable address of the subnetwork.
2. Using the IP addresses calculated, configure the devices with the appropriate information.

Subnet requirements from Table 1:

1. **JERUNG LAN1:** Requires 20 hosts $\rightarrow 2^5 = 32$ addresses
 Subnet mask: **/27**
 Usable IPs: $32 - 2 = 30$
2. **JERUNG LAN2:** Requires 30 hosts $\rightarrow 2^5 = 32$ addresses
 Subnet mask: **/27**
 Usable IPs: $32 - 2 = 30$
3. **SHARK LAN:** Requires 60 hosts $\rightarrow 2^6 = 64$ addresses
 Subnet mask: **/26**
 Usable IPs: $64 - 2 = 62$
4. **TIBURON LAN:** Requires 50 hosts $\rightarrow 2^6 = 64$ addresses
 Subnet mask: **/26**
 Usable IPs: $64 - 2 = 62$
5. **Router-to-Router Links:** Each link requires 2 hosts $\rightarrow 2^2 = 4$ addresses
 Subnet mask: **/30**
 Usable IPs: $4 - 2 = 2$

Device	Subnetwork	Total IPs	Usable IPs
JERUNG	LAN1	172.18.110.0 – 172.18.110.31	172.18.110.1 – 172.18.0.30
	LAN2	172.18.110.32 – 172.18.110.63	172.18.110.33 – 172.18.0.62
SHARK	LAN	172.18.110.64 – 172.18.110.127	172.18.110.65 – 172.18.110.126
TIBURON	LAN	172.18.110.128 – 172.18.110.191	172.168.110.129 – 172.18.110.190
CONNECTIONS	JERUNG-SHARK	172.18.110.192 – 172.18.110.195	172.18.110.193 – 172.18.110.194
	JERUNG-TIBURON	172.18.110.196 – 172.18.110.199	172.18.110.197 – 172.18.110.198
	SHARK-TIBURON	172.18.110.200 – 172.18.110.203	172.18.110.201 – 172.18.110.202

Device	Subnetwork	Prefix	Subnet Mask
JERUNG	LAN1	/27	11111111.11111111.11111111.11100000 = 255.255.255.224
	LAN2	/27	11111111.11111111.11111111.11111100 = 255.255.255.252
SHARK	LAN	/26	11111111.11111111.11111111.11000000 = 255.255.255.192
TIBURON	LAN	/26	11111111.11111111.11111111.11000000 = 255.255.255.192
Connections	JERUNG-SHARK	/30	11111111.11111111.11111111.11111100 = 255.255.255.252
	JERUNG-TIBURON	/30	11111111.11111111.11111111.11111100 = 255.255.255.252
	SHARK-TIBURON	/30	11111111.11111111.11111111.11111100 = 255.255.255.252

Task 2 – Routing Table

1. Paste the current routing table of each router here. There are 2 ways to this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers.

- a. CLI

- i. Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
       C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
       L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
```

Figure A

- b. PT tool

- i. Click on the magnifying glass (shown in Figure B), then click on a router, then choose Routing Table. A sample of the result is shown in Figure C.

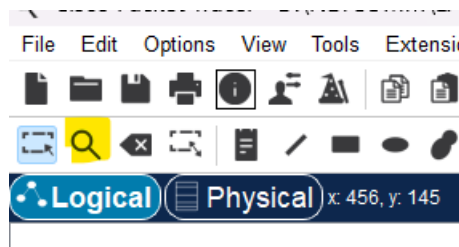


Figure B

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0

Figure C

JERUNG:

CLI

```
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
C    172.18.110.0/27 is directly connected, GigabitEthernet0/0
L    172.18.110.1/32 is directly connected, GigabitEthernet0/0
C    172.18.110.32/27 is directly connected, GigabitEthernet0/1
L    172.18.110.33/32 is directly connected, GigabitEthernet0/1
C    172.18.110.192/30 is directly connected, Serial0/0/0
L    172.18.110.193/32 is directly connected, Serial0/0/0
C    172.18.110.196/30 is directly connected, Serial0/0/1
L    172.18.110.197/32 is directly connected, Serial0/0/1
```

PT Tool

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0

SHARK:

CLI

```
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C    172.18.110.64/26 is directly connected, GigabitEthernet0/0
L    172.18.110.65/32 is directly connected, GigabitEthernet0/0
C    172.18.110.192/30 is directly connected, Serial0/0/0
L    172.18.110.194/32 is directly connected, Serial0/0/0
C    172.18.110.200/30 is directly connected, Serial0/0/1
L    172.18.110.201/32 is directly connected, Serial0/0/1
```

PT Tool

HKMS12425

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

TIBURON:

CLI

```
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C    172.18.110.128/26 is directly connected, GigabitEthernet0/0
L    172.18.110.129/32 is directly connected, GigabitEthernet0/0
C    172.18.110.196/30 is directly connected, Serial0/1/1
L    172.18.110.198/32 is directly connected, Serial0/1/1
C    172.18.110.200/30 is directly connected, Serial0/1/0
L    172.18.110.202/32 is directly connected, Serial0/1/0
```

PT Tool

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

2. Try to ping PC2 from PC1, paste the results here.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.62: bytes=32 time=6ms TTL=128
Reply from 172.18.110.62: bytes=32 time<1ms TTL=128
Reply from 172.18.110.62: bytes=32 time=3ms TTL=128
Reply from 172.18.110.62: bytes=32 time=6ms TTL=128

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 3ms
```

3. Try to ping PC4 from PC1, paste the results here.

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

4. Explain the reason(s) behind the results.
It is because the subnet for PC1 and PC4 are different so that they cannot communicate directly with each other. Next, they are connected to different routers as the result shows “Destination host unreachable”.
5. What needs to be done to ensure all PCs can ping each other successfully?
To ensure all PCs can ping each other successfully, the following steps need to be implemented:
- **Configure IP Addressing:** Assign valid IP addresses, subnet masks, and default gateways to all PCs based on their respective subnets.
 - **Enable All Router Interfaces:** Use the no shutdown command on all router interfaces to ensure they are active.
 - **Configure Routing Protocols:** Implement a routing protocol, such as RIP (Routing Information Protocol), on all routers to ensure that routes to all subnets are shared and known across the network.

Task 2 – Routing Configuration

1. Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG. This is done to show changes to the routing table as configurations are made.

Routing Table for TIBURON					Routing Table for JERUNG				
Type	Network	Port	Next Hop IP	Metric	Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0	C	172.18.110.128/27	GigabitEthernet0/1	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0	L	172.18.110.129/32	GigabitEthernet0/1	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0	C	172.18.110.160/27	GigabitEthernet0/0	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0	L	172.18.110.161/32	GigabitEthernet0/0	---	0/0

Figure D

2. In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```

JERUNG#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#

```

Figure E

- a. What can you say about the addresses used in the 'network' instructions in Figure E?
The network instructions tell the RIP to advertise and include all the interfaces within the subnets.

network 172.18.110.128 – include all interface within 172.18.110.128 subnet

network 172.18.110.160 – include all interface within 172.18.110.160 subnet

network 172.18.110.192 – include all interface within 172.18.110.192 subnet

network 172.18.110.196 – include all interface within 172.18.110.196 subnet

3. Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```

TIBURON (config-router) #network 172.18.110.64
TIBURON (config-router) #network 172.18.110.196
TIBURON (config-router) #network 172.18.110.200

```

4. As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here.

JERUNG

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0
R	172.18.110.64/26	Serial0/0/0	172.18.110.194	120/1
R	172.18.110.128/26	Serial0/0/1	172.18.110.198	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0
R	172.18.110.200/30	Serial0/0/1	172.18.110.198	120/1
R	172.18.110.200/30	Serial0/0/0	172.18.110.194	120/1

TIBURON

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.64/26	Serial0/1/0	172.18.110.201	120/1
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

- a. What are the changes seen in TIBURON?

Addition of RIP route for subnet 172.18.110.0/27, 172.18.110.32/27, 172.18.110.64/26 and 172.18.119.192/30. These routes, learned from another router via RIP, are now part of the TIBURON routing table, enabling the router to forward traffic to these subnets.

- b. What are the Networks with type R in TIBURON and JERUNG?

TIBURON – 172.18.110.0/27, 172.18.110.32/27, 172.18.119.64/26

JERUNG – 172.18.110.64/26, 172.18.110.128/26

- c. Ping PC3 from PC1. Was it successful?

Yes

```
C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.190: bytes=32 time=7ms TTL=126
Reply from 172.18.110.190: bytes=32 time=5ms TTL=126
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 7ms, Average = 4ms
```

- d. Ping PC4 from PC1. Was it successful?

Yes

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.126: bytes=32 time=13ms TTL=126
Reply from 172.18.110.126: bytes=32 time=23ms TTL=126
Reply from 172.18.110.126: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 23ms, Average = 12ms
```

- e. Explain the reasons for your answer.

This is because RIP enables routers to dynamically exchange routing information, facilitating end-to-end communication between PCs across different subnets.

Additionally, the correct configuration of IP addresses, subnet masks, and default gateways ensures that packets are properly routed to their intended destinations.

- f. Continue with configuration of RIP in SHARK. Paste your configurations here.

```
SHARK#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SHARK(config)#router rip
SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110.64
SHARK(config-router)#network 172.18.110.192
SHARK(config-router)#network 172.18.110.200
SHARK(config-router)#no auto-summary
```

- g. Open router SHARK's routing table and paste here.

```

R    172.18.110.0/27    Serial0/0/0    172.18.110.193 120/1

R    172.18.110.32/27   Serial0/0/0    172.18.110.193 120/1

C    172.18.110.64/26   GigabitEthernet0/0 ---          0/0

L    172.18.110.65/32   GigabitEthernet0/0 ---          0/0

R    172.18.110.128/26  Serial0/0/1    172.18.110.202 120/1

C    172.18.110.192/30  Serial0/0/0    ---          0/0

L    172.18.110.194/32  Serial0/0/0    ---          0/0

R    172.18.110.196/30  Serial0/0/0    172.18.110.193 120/1

R    172.18.110.196/30  Serial0/0/1    172.18.110.202 120/1

C    172.18.110.200/30  Serial0/0/1    ---          0/0

L    172.18.110.201/32  Serial0/0/1    ---          0/0

```

- h. Try to ping from PC4 to all other PCs in the topology. *Note: Try to ping at least twice to get best results.

PC	Ping result
1	<pre> C:\>ping 172.18.110.30 Pinging 172.18.110.30 with 32 bytes of data: Reply from 172.18.110.30: bytes=32 time=2ms TTL=128 Reply from 172.18.110.30: bytes=32 time=1ms TTL=128 Reply from 172.18.110.30: bytes=32 time=4ms TTL=128 Reply from 172.18.110.30: bytes=32 time=4ms TTL=128 Ping statistics for 172.18.110.30: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 4ms, Average = 2ms C:\>ping 172.18.110.30 Pinging 172.18.110.30 with 32 bytes of data: Reply from 172.18.110.30: bytes=32 time=4ms TTL=128 Reply from 172.18.110.30: bytes=32 time=2ms TTL=128 Reply from 172.18.110.30: bytes=32 time=4ms TTL=128 Reply from 172.18.110.30: bytes=32 time=4ms TTL=128 Ping statistics for 172.18.110.30: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 2ms, Maximum = 4ms, Average = 3ms </pre>

2	<pre> C:\>ping 172.18.110.33 Pinging 172.18.110.33 with 32 bytes of data: Reply from 172.18.110.33: bytes=32 time<1ms TTL=255 Reply from 172.18.110.33: bytes=32 time<1ms TTL=255 Reply from 172.18.110.33: bytes=32 time<1ms TTL=255 Reply from 172.18.110.33: bytes=32 time<1ms TTL=255 Ping statistics for 172.18.110.33: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms C:\>ping 172.18.110.33 Pinging 172.18.110.33 with 32 bytes of data: Reply from 172.18.110.33: bytes=32 time<1ms TTL=255 Reply from 172.18.110.33: bytes=32 time<1ms TTL=255 Reply from 172.18.110.33: bytes=32 time<1ms TTL=255 Reply from 172.18.110.33: bytes=32 time<1ms TTL=255 Ping statistics for 172.18.110.33: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms </pre>
3	<pre> C:\>ping 172.18.110.190 Pinging 172.18.110.190 with 32 bytes of data: Reply from 172.18.110.190: bytes=32 time<1ms TTL=128 Reply from 172.18.110.190: bytes=32 time<1ms TTL=128 Reply from 172.18.110.190: bytes=32 time=2ms TTL=128 Reply from 172.18.110.190: bytes=32 time=4ms TTL=128 Ping statistics for 172.18.110.190: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 4ms, Average = 1ms C:\>ping 172.18.110.190 Pinging 172.18.110.190 with 32 bytes of data: Reply from 172.18.110.190: bytes=32 time=4ms TTL=128 Reply from 172.18.110.190: bytes=32 time<1ms TTL=128 Reply from 172.18.110.190: bytes=32 time=5ms TTL=128 Reply from 172.18.110.190: bytes=32 time=2ms TTL=128 Ping statistics for 172.18.110.190: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 5ms, Average = 2ms </pre>

Task 3 – Routing Update

1. Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.

- a. This means that the subnet has changed. Find the new Network address of this subnet.
Network Address is: 192.168.1.0/24
- b. As this change happens, PC3 must also have different IP address, subnet mask and gateway address. What will it be?

PC3	Info
IP address	192.168.1.2
Subnet Mask	255.255.255.0
Gateway Address	192.168.1.1

- c. After this change, can PC4 and PC1 ping PC3?
No, because they are on different gateway. Routing needs to be done between 172.18.110.0/27, 172.18.110.0/26 and 192.168.1.0/24 in order to communicate with each other.
- d. Copy and paste the routing tables for both SHARK and TIBURON here.

SHARK

```

R    172.18.110.0/27  Serial0/0/0      172.18.110.193 120/1
R    172.18.110.32/27 Serial0/0/0      172.18.110.193 120/1
C    172.18.110.64/26 GigabitEthernet0/0 ---      0/0
L    172.18.110.65/32 GigabitEthernet0/0 ---      0/0
C    172.18.110.192/30 Serial0/0/0      ---      0/0
L    172.18.110.194/32 Serial0/0/0      ---      0/0
R    172.18.110.196/30 Serial0/0/0      172.18.110.193 120/1
R    172.18.110.196/30 Serial0/0/1      172.18.110.202 120/1
C    172.18.110.200/30 Serial0/0/1      ---      0/0
L    172.18.110.201/32 Serial0/0/1      ---      0/0

```

TIBURON

R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.64/26	Serial0/1/0	172.18.110.201	120/1
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0
C	192.168.1.0/26	GigabitEthernet0/0	---	0/0
L	192.168.1.1/32	GigabitEthernet0/0	---	0/0

- e. Referring to the routing table, explain your findings.
 JERUNG's routing table, contains RIP-learned (R) routes for networks such as 172.18.110.0/27, 172.18.110.32/27, and 172.18.110.64/26.
 TIBURON's routing table contains RIP-learned (R) routes for networks 172.18.110.0/27, 172.18.110.32/27, and 172.18.110.64/26.
 Both routers rely on RIP, which shows R (RIP-learn) in the routing table.
- f. What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?
 Ensure that the TIBURON have the RIP routes for other subnets.
 Make sure that the PCs have valid IP address, subnet masks and gateways.
- g. Show your configurations in TIBURON to ensure end-to-end connectivity.
- ```
TIBURON#
%SYS-5-CONFIG_I: Configured from console by console
configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
TIBURON(config)#router rip
TIBURON(config-router)#version 2
TIBURON(config-router)#network 192.168.1.0
TIBURON(config-router)#no auto-summary
TIBURON(config-router)#exit
```
- h. To ensure end-to-end connectivity, ping to all the PCs from PC3.



| PC | Ping result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | <pre> C:\&gt;ping 172.18.110.30  Pinging 172.18.110.30 with 32 bytes of data:  Reply from 172.18.110.30: bytes=32 time=5ms TTL=128 Reply from 172.18.110.30: bytes=32 time=4ms TTL=128 Reply from 172.18.110.30: bytes=32 time=5ms TTL=128 Reply from 172.18.110.30: bytes=32 time=4ms TTL=128  Ping statistics for 172.18.110.30:     Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),     Approximate round trip times in milli-seconds:         Minimum = 4ms, Maximum = 5ms, Average = 4ms  C:\&gt;ping 172.18.110.30  Pinging 172.18.110.30 with 32 bytes of data:  Reply from 172.18.110.30: bytes=32 time=5ms TTL=128 Reply from 172.18.110.30: bytes=32 time=2ms TTL=128 Reply from 172.18.110.30: bytes=32 time=2ms TTL=128 Reply from 172.18.110.30: bytes=32 time=2ms TTL=128  Ping statistics for 172.18.110.30:     Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),     Approximate round trip times in milli-seconds:         Minimum = 2ms, Maximum = 5ms, Average = 2ms </pre>      |
| 2  | <pre> C:\&gt;ping 172.18.110.62  Pinging 172.18.110.62 with 32 bytes of data:  Reply from 172.18.110.62: bytes=32 time=6ms TTL=128 Reply from 172.18.110.62: bytes=32 time=2ms TTL=128 Reply from 172.18.110.62: bytes=32 time=16ms TTL=128 Reply from 172.18.110.62: bytes=32 time=6ms TTL=128  Ping statistics for 172.18.110.62:     Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),     Approximate round trip times in milli-seconds:         Minimum = 2ms, Maximum = 16ms, Average = 7ms  C:\&gt;ping 172.18.110.62  Pinging 172.18.110.62 with 32 bytes of data:  Reply from 172.18.110.62: bytes=32 time&lt;1ms TTL=128 Reply from 172.18.110.62: bytes=32 time=4ms TTL=128 Reply from 172.18.110.62: bytes=32 time=3ms TTL=128 Reply from 172.18.110.62: bytes=32 time=2ms TTL=128  Ping statistics for 172.18.110.62:     Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),     Approximate round trip times in milli-seconds:         Minimum = 0ms, Maximum = 4ms, Average = 2ms </pre> |



|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 | <pre>C:\&gt;ping 172.18.110.126  Pinging 172.18.110.126 with 32 bytes of data:  Reply from 172.18.110.126: bytes=32 time=7ms TTL=128 Reply from 172.18.110.126: bytes=32 time=4ms TTL=128 Reply from 172.18.110.126: bytes=32 time=4ms TTL=128 Reply from 172.18.110.126: bytes=32 time=2ms TTL=128  Ping statistics for 172.18.110.126:     Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds:     Minimum = 2ms, Maximum = 7ms, Average = 4ms  C:\&gt;ping 172.18.110.126  Pinging 172.18.110.126 with 32 bytes of data:  Reply from 172.18.110.126: bytes=32 time=7ms TTL=128 Reply from 172.18.110.126: bytes=32 time&lt;1ms TTL=128 Reply from 172.18.110.126: bytes=32 time&lt;1ms TTL=128 Reply from 172.18.110.126: bytes=32 time=5ms TTL=128  Ping statistics for 172.18.110.126:     Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds:     Minimum = 0ms, Maximum = 7ms, Average = 3ms</pre> |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## ***REFLECTION***

What have you learned in this task?

In this lab, I learned the importance of proper subnetting, IP configuration, and the role of routing protocols like RIP in enabling communication between devices in different subnets. Configuring router interfaces, enabling them with ‘no shutdown’, and verifying connectivity using commands like ‘show ip route’ helped me understand how routers dynamically exchange routing information to establish end-to-end communication. This exercise also highlighted the need for accurate IP address assignments, subnet masks, and default gateways to ensure successful packet delivery across the network.